

HAMZA ALTAF

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PROFILE

Computer Science (Cybersecurity) undergraduate at Brunel University London, achieving First Class standing in Year 1. Passionate backend engineer with production experience in Java (OOP), JDBC, and SQL, and a strong understanding of RESTful architecture, data integrity, and system workflow design. Currently studying Operating Systems & Networks and Software Engineering, deepening knowledge in distributed systems, concurrency, process scheduling, and scalable backend design. Actively interested in the global gaming industry and the specific technical challenges of building high-availability, low-latency backend services for large-scale unified account platforms. Proficient user of AI coding tools including Claude and GitHub Copilot to accelerate development, improve code quality, and explore new approaches.

TECHNICAL SKILLS

Backend & Languages

- Java (OOP, JDBC) — primary production language
- Python (familiar), SQL, C# (familiar)
- RESTful architecture & API design patterns
- CRUD operations, data integrity, system workflows
- Algorithms & data structures

Systems & Infrastructure

- Relational DB design — SQL queries, JOINS, indexing
- Operating Systems & Networks (current module)
- Authentication & access control (IAM)
- Cloud security & endpoint security (MS cert)
- Vulnerability management & secure system design

Frontend & Tooling

- JavaScript, HTML, CSS, React
- Git/GitHub — version control & team collaboration
- Jira — Agile sprint & backlog management
- AI coding tools: Claude, GitHub Copilot
- Unit testing & technical documentation

Professional Skills

- Agile & Scrum methodology (certified, practised)
- Cross-functional team collaboration & code review
- Analytical debugging & root cause analysis
- Clear written & verbal communication
- Fast adaptation to new languages & frameworks

TECHNICAL EXPERIENCE

Software Developer Intern

Jun – Jul 2025

Daska City Housing Scheme, Daska, Pakistan

- Designed and developed the complete backend of a Real Estate Management Dashboard using Java, JDBC, and SQL — architecting the data layer from scratch, implementing end-to-end CRUD operations, and building reliable system workflows for a live production environment used by non-technical staff daily.
- Engineered relational SQL database schemas with careful attention to normalisation, join performance, and data integrity; wrote complex multi-table queries with null-safety handling and validation logic to ensure consistent, accurate output across all user-facing reports.
- Identified and resolved a critical pre-launch data integrity bug: a SQL JOIN was silently returning incorrect results for records with null optional fields; traced root cause through the Java JDBC layer, implemented a robust null-handling fix, and added regression test coverage to prevent recurrence.
- Worked across the full software development lifecycle — from initial requirements gathering with non-technical business stakeholders through system design, iterative Agile development, QA, deployment, and post-launch user training.
- Produced comprehensive technical documentation including system architecture overviews, database schema diagrams, API workflow descriptions, and end-user operation guides to support long-term operational use and future development.

EDUCATION

BSc Computer Science (Cybersecurity)

Sep 2024 – Jun 2027

Brunel University London | Final Year: First Class Standing | Expected Jun 2027

- Completed: Introductory Programming (A*), Software Design (A), Data & Information (B+), Logic & Computation (A)
- Currently studying: Databases, Software Engineering, Operating Systems & Networks — building direct, applied knowledge in backend architecture, network protocols, concurrency models, and scalable distributed system design
- Final-year focus: deepening practical backend engineering experience ahead of graduate-level study

Foundation Degree in Science

Jan – Aug 2024

Kingston University, Kingston upon Thames

- Pure Mathematics (82%), Core Physics (84%), Academic English (75%) — rigorous analytical and quantitative foundation directly applicable to algorithm design, data modelling, and systems reasoning

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Microsoft Cybersecurity Analyst — Professional Certificate

In Progress

- Cloud security architecture, identity and access management (IAM), endpoint security, vulnerability management, and incident response — building a strong understanding of the security considerations underpinning high-availability, distributed backend service design
- Covers cloud-native security patterns, threat modelling, and compliance frameworks relevant to globally deployed gaming platforms and account systems

- Scrum ceremonies, sprint planning, backlog management, velocity tracking, and iterative delivery in cross-functional engineering teams — certified and practised across multiple live projects

PORTFOLIO

SafeCircle — Disaster Support Platform

GitHub: github.com/HamzaAltaf-799

Oct 2025 – Present | Active

React • JavaScript • HTML/CSS • GitHub • Jira

- Architected multi-role authentication system and priority-based alerting logic routing requests by user role and incident severity — backend-facing access control design directly relevant to unified account platforms like Level Infinite PASS
- Scrum Master: led sprint planning, backlog refinement, retrospectives; maintained Jira board, user stories, status updates, and full Agile documentation across the project lifecycle
- Designed and built the Crisis Assistance Page with real-time user-role-based routing, guardian alert workflows, and accessibility features for multi-role user support
- Conducted requirements analysis and user acceptance testing against functional and non-functional criteria; evaluated alternative platform architectures and delivered evidence-based design recommendations
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Real Estate Management Dashboard

Jun – Jul 2025 | Deployed to production

Java • JDBC • SQL • OOP • Agile

- Full backend system built for a live production environment: architected data layer, implemented CRUD operations, and engineered end-to-end business logic entirely in Java and SQL
- Designed relational database schema with normalisation, complex JOIN queries, and null-safety validation to ensure data consistency across all user-facing reports
- Resolved a critical data integrity bug pre-launch: traced a silent SQL JOIN failure through the JDBC layer and implemented a robust fix with proper null-handling and regression coverage
- Produced full technical architecture documentation, database schema diagrams, and plain-language end-user guides deployed to non-technical operational staff
- Delivered across the full development lifecycle using Agile: requirements, design, development, QA, deployment, and post-launch user training

SwiftBot Shape Detection System

May 2025 | Completed with distinction

Java • SwiftBot API • Algorithms • Hardware Integration

- Designed and implemented a Java application integrating the SwiftBot hardware API for real-time shape detection, classification, and automated drawing — applying algorithms and data structures for efficient, modular processing under hardware constraints
- Debugged hardware-software interface issues at the API boundary, diagnosing communication failures between the Java application layer and the physical device — building experience in low-level system integration and real-time data processing
- Applied OOP design principles for clean, modular, extensible architecture; documented system design decisions and algorithm choices clearly for peer review and technical assessment
- Delivered a live technical demonstration to assessors in an Agile team environment, presenting the system end-to-end and fielding technical questions about design decisions

Simon Swift — Pattern Recognition Game

Mar 2025 | Completed

Java • Event-Driven Programming • OOP • GitHub

- Designed and implemented scalable pattern recognition logic and adaptive difficulty algorithms using event-driven programming in Java, with careful attention to game state management and input handling
- Applied OOP principles throughout: clean class hierarchy, separation of concerns, and extensible architecture allowing new difficulty levels and game modes to be added without refactoring core logic
- Collaborated in a shared GitHub repository with branch-based workflow, peer code review, and structured commit history — practising real engineering team collaboration conventions
- Produced comprehensive technical documentation and testing evidence including unit test results, edge case analysis, and a post-project technical reflection for assessment and peer review

ADDITIONAL EXPERIENCE

Retail Assistant — John Lewis Partnership (Waitrose)

Current

- Delivers consistent, high-quality service in a fast-paced, high-footfall environment, developing strong communication under pressure, task prioritisation, and stakeholder relationship management skills — professional attributes directly transferable to cross-functional team collaboration and user-facing support in a global engineering organisation.
- Regularly handles complex or sensitive customer situations requiring empathy, clear communication, and rapid problem resolution — building the interpersonal skills essential for working effectively across teams in a diverse, international company like Level Infinite.

INTERESTS & ADDITIONAL INFORMATION

Avid gamer with a genuine interest in the technical architecture of online gaming platforms — particularly backend systems for real-time multiplayer, account management, and cross-regional service delivery. Follows developments in cloud-native technologies, distributed systems, and the evolution of game backend infrastructure. Active contributor to group coding projects on GitHub. Committed to continuous learning: currently exploring Go fundamentals and distributed systems concepts alongside university studies to build towards the backend architecture skills required in this role.